EECE 571Q

Lecture 10: Beneath the surface (part 2)

Thursday 11 June 2026

Announcements

- Project proposal due tomorrow Friday 12 June 23:59
  - please advise of any lecture date constraints
  - I will try to order lectures based on content
- Exam reference sheet info available (due Thursday 18 June)
- Class on Zoom next week
- Quiz 5 on Tuesday

Last time

We introduced the **surface code** (2D planar code), and saw some examples of operations that can be performed on it.

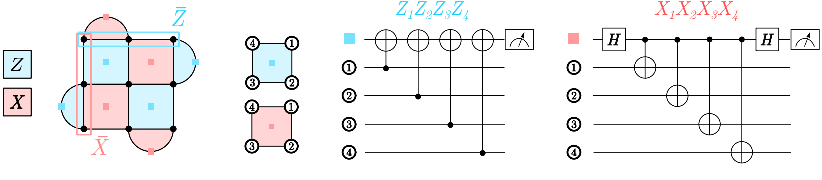
Last time

We briefly discussed code distance and how errors manifest in the syndrome of stabilizer measurements.

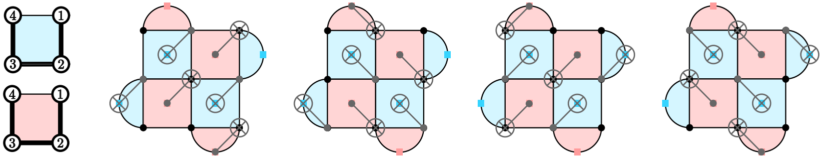
Learning outcomes

1. Construct circuits for surface code syndrome measurement
2. Describe general decoding strategies, and specific decoding strategies for surface codes
3. Carry out simulations in stim to estimate the surface code (pseudo)threshold

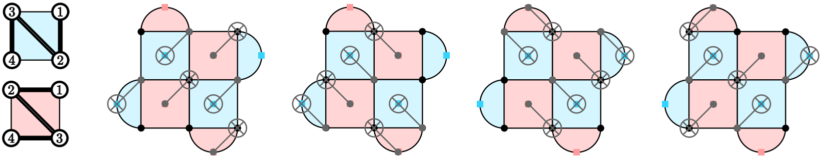
Syndrome measurement



Syndrome measurement



Syndrome measurement



Syndromes and cosets

When we studied the 9-qubit code (L4), we found multiple errors that produced the same syndrome.

Syndromes and decoding

Exercise: What are the error syndromes for Y_1 and $X_0 Z_1 X_2$?

Name	Operator
g_1	$ZZIIIIIII$
g_2	$IZZIIIIII$
g_3	$IIIZZIIII$
g_4	$IIIIZZIII$
g_5	$IIIIIIZZI$
g_6	$IIIIIIIZZ$
g_7	$XXXXXXIII$
g_8	$IIIXXXXXX$
$\bar{Z}$	$XXXXXXXXX$
$\bar{X}$	$ZZZZZZZZZ$

Screenshot: Nielsen & Chuang, chapter 10.5.6.

Syndromes and cosets

Why this happens: cosets of $N(S)$ w.r.t. the Pauli group.

Gottesman Proposition 3.8 (rephrased)

Proposition: Two errors, E and F, are in the same coset of $N(S)$ if they have the same error syndrome.

Example: $S = \{ZZI, IZZ\}$. same coset means $F = EN$, $N \in N(S)$.
$N(S) = \{I, ZZI, IZZ, ZIZ, ZZZ, ZII, IZI, IIZ, XXX, YYY, YXY, XYX, YXX, XYY, XXY, YYX\}$

Syndromes and cosets

Similarly: consider cosets of S in $N(S)$.

Gottesman Proposition 3.10 (rephrased)

Proposition: Let S be a stabilizer, and consider $N_1, N_2 \in N(S)$. N_1 and N_2 are in the same coset of S iff, for all codewords $|\psi\rangle$, $N_1|\psi\rangle = N_2|\psi\rangle$.

Proof: $|\psi\rangle = N_1^\dagger N_2|\psi\rangle$; thus, $M = N_1^\dagger N_2 \in S$.
the consequence is since each N maps code words to other code words, it is a logical operator. And every other logical operator in its coset is just a different way of realizing the same logical operation. This is why on the surface code we can choose the logical operators to be any column/row: they are in the same coset $N(S) = \{I, ZZI, IZZ, ZIZ, ZZZ, ZII, IZI, IIZ, XXX, YYY, YXY, XYX, YXX, XYY, XXY, YYX\}$. Each of the four cosets is associated with a different error syndrome associated to a class of logical errors

Maximum likelihood decoding

Decoding is the process of interpreting the syndrome measurement and making a decision about which error actually occurred.

Initial idea: associate each error syndrome with the **lowest-weight representative** of each coset of $N(S)$.

Better option: maximum likelihood decoding.

Proposition 3.13. *Let C_s be the coset of $\hat{\mathsf{N}}(\mathsf{S})$ with error syndrome s. $\hat{Q}_s\hat{\mathsf{S}}$ is the coset of $\hat{\mathsf{S}}$ containing the canonical error $\hat{Q}_s$ with syndrome s, so $\hat{Q}_s\hat{\mathsf{S}} \subseteq C_s$.*

a) The probability of error syndrome s (regardless of whether we correctly identify the error) is

$$p_s = \sum_{\hat{P} \in C_s} p_P. \tag{3.22}$$

b) The probability of having syndrome s and no logical error (i.e., error correction is successful) is

$$p_{s,\mathrm{OK}} = \sum_{\hat{P} \in \hat{Q}_s\hat{\mathsf{S}}} p_P. \tag{3.23}$$

c) The total probability of successfully decoding the state is

$$p_{\mathrm{OK}} = \sum_s \sum_{\hat{P} \in \hat{Q}_s\hat{\mathsf{S}}} p_P. \tag{3.24}$$

Gottesman Proposition 3.13

Maximum likelihood decoding

Example: bit flip code, measure syndrome 10. X, Y, Z error occurs independently on each qubit with probability $p/3$.

Cosets of $N(S)$:

$(III)00 : \{III, ZZI, IZZ, ZIZ, ZZZ, ZII, IZI, IIZ, XXX, YYY, YXY, XYX, YXX, XYY, XXY, YYX\}$

$(IIX)01 : \{IIX, ZZX, IZY, ZIY, ZZY, ZIX, IZX, IIY, XXI, YYZ, YXZ, XYI, YXI, XYZ, XXZ, YYI\}$

$(XII)10 : \{XII, YZI, XZZ, YIZ, YZZ, YII, XZI, XIZ, IXX, ZYY, ZXY, IYX, ZXX, IYY, IXY, ZYX\}$

$(IXI)11 : \{IXI, ZYI, IYZ, ZXZ, ZYZ, ZXI, IYI, IXZ, XIX, YZY, YIY, XZX, YIX, XZY, XIY, YZX\}$

Probability of syndromes 01, 10, 11: $\frac{2}{3}p(1-p)^2 + \frac{8}{9}p^2(1-p) + \frac{6}{27}p^3$
Probability of syndrome 00: $(1-p)^3 + p(1-p)^2 + \frac{1}{3}p^2(1-p) + \frac{1}{3}p^3$

Maximum likelihood decoding

Good choice: coset representative XII:

$(XII)10$: $\{\mathbf{XII}, \mathbf{YZI}, \mathbf{XZZ}, \mathbf{YIZ}, YZZ, YII, XZI, XIZ, IXX, ZYY, ZXY, IYX, ZXX, IYY, IXY, ZYX\}$

Probability of syndrome 10 and no logical error:
$\frac{p}{3}(1-p)^2 + \frac{2}{9}p^2(1-p) + \frac{p^3}{27}$

For *each symdrome*, choose coset representative that *maximizes* probability of no logical error if applied as correction.

Bad choice: ZXX. Coset of S is $\{XZZ, IYX, ZYY, IXY\}$.

Minimum weight perfect matching

Determining the optimal choice and decoding strategy in this way is computationally challenging.

On the surface code, we can leverage structure of how errors manifest to take an alternative approach.

highlight an example that shows the hardest case to distinguish between is chains of distance $d - 1/2$ and $d + 1/2$, which is where the scaling on threshold plots comes from.

Minimum weight perfect matching

We must also consider how measurement errors propagate in time.
Perform d rounds of syndrome measurements to identify them.

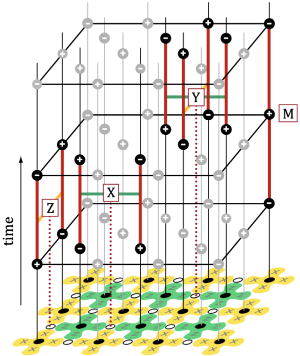


Image: Fowler et al., (2012), *Surface codes: Towards practical large-scale quantum computation*, PRA86 (3) 032324.

Stim simulations

Let's use stim to perform some experiments with various error models and estimate the threshold of the surface code.

Next time

Next class:

- Quiz 5
- Magic states: injection (gate teleportation) and distillation

Action items:

1. Work on and submit project proposal
2. Start preparing exam reference sheet

Recommended reading for this / next class:

- Gottesman Ch. 3.4 and 13; N & C 10.6
- Fowler, Mariatoni, Martini, and Cleland (2012), *Surface codes: Towards practical large-scale quantum computation*, Phys. Rev. A 86 (3) 032324.
- Horsman, Fowler, Devitt, and van Meter (2012), *Surface code quantum computing by lattice surgery*, NJP 14 123011.
- PennyLane tutorial on magic states
- PennyLane tutorial on state distillation